

# DIGITAL OUTLET

REVOLUTIONIZING INDUSTRIES,  
TRANSFORMING LIVES



## CYBER FUTURE

HOW TECHNOLOGY IS SAFEGUARDING  
THE DIGITAL WORLD

## GREEN TECH

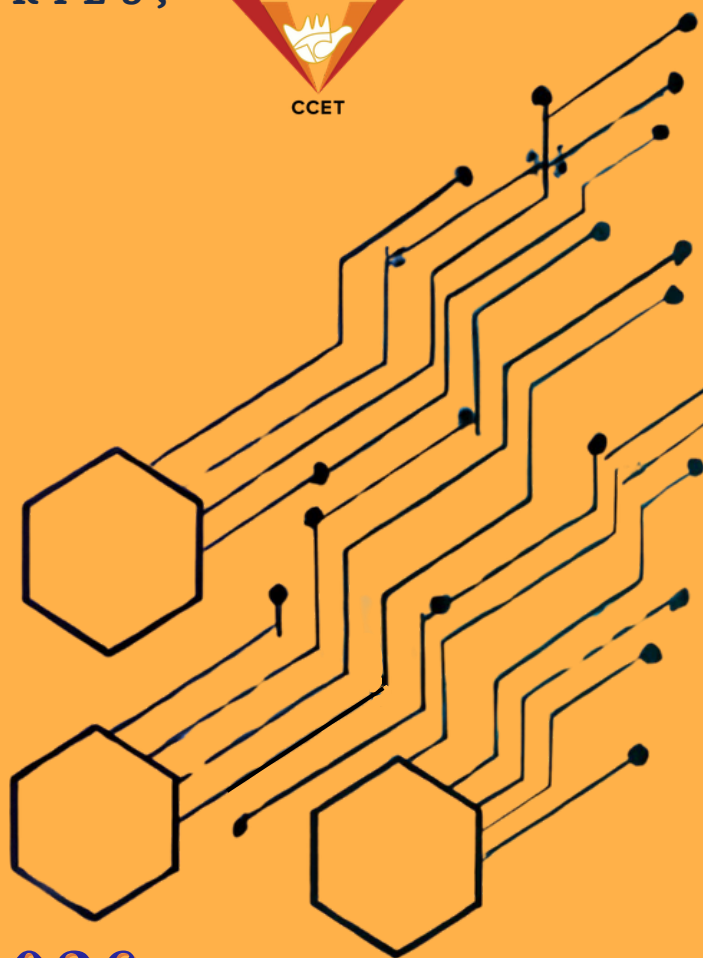
SUSTAINABILITY MEETS  
SMART SOLUTIONS

## SPACE TECH

SMART GADGETS TO SIMPLIFY  
YOUR LIFE

## CYBERSECURITY 2026

ARE WE PREPARED?



PUBLISHED BY-  
CCET ACM STUDENT  
CHAPTER  
CCET, DEGREE WING,  
SECTOR 26 CHANDIGARH



## 1. Journal Articles

- **Ensemble deep learning and EfficientNet for accurate diagnosis of diabetic retinopathy**

*Authors - Lakshay Arora, Sunil K Singh, Sudhakar Kumar, Hardik Gupta, Wadee Alhalabi, Varsha Arya, Shavi Bansal, Kwok Tai Chui, Brij B Gupta*

- **AI-Based Identification of Drug Use and Overdose Signals on Social Media**

*Authors - Sudhakar Kumar, Sunil K. Singh, Ryanveer Singh, Pooja Rai, Varsha Arya, Brij B. Gupta*

- **Mobile cloud computing and sustainable development: Opportunities, challenges, and future directions**

*Authors - Francisco Peñalvo, Sunil Singh, Sudhakar Kumar, Akash Sharma, Anureet Chhabra, Varsha Arya, Akshat Gaurav*

## 2. Conference Papers

- **Robust Multi-Modal RF Fingerprinting in Intelligent Reality Using Federated Learning**

*Authors - Sudhakar Kumar, Sunil K Singh, Sunil Singh, Shivam Nayal, Varsha Arya, Kwok Tai Chui, Brij B Gupta*

- **A Novel Approach for Evaluating Spatial-Temporal Synergy in Hybrid CNN-RNN and Vision Transformer Architectures**

*Authors - Viren Passi, Sudhakar Kumar, Sunil K Singh, Shreya Verma, Varsha Arya, Valerie Tang, Brij B Gupta, Kwok Tai Chui*



- **Real-Time Privacy Guardianship in XR Metaverses via AI-Driven Behavioral Analytics**

*Authors - Ritika Kalia, Sudhakar Kumar, Sunil K Singh, Samriti Sharma, Kwok Tai Chui, Brij B Gupta*

- **AI-Driven Deep Learning for Real-Time DDoS Detection in Software-Defined Networks**

*Authors - Sudhakar Kumar, Sunil K Singh, Rakesh Kumar, Satyam Kumar Sainy, Anand Kumar Yadav, Shabeg Singh Gill*

- **Ensemble-Driven Malware Detection with Anomaly Override Capabilities**

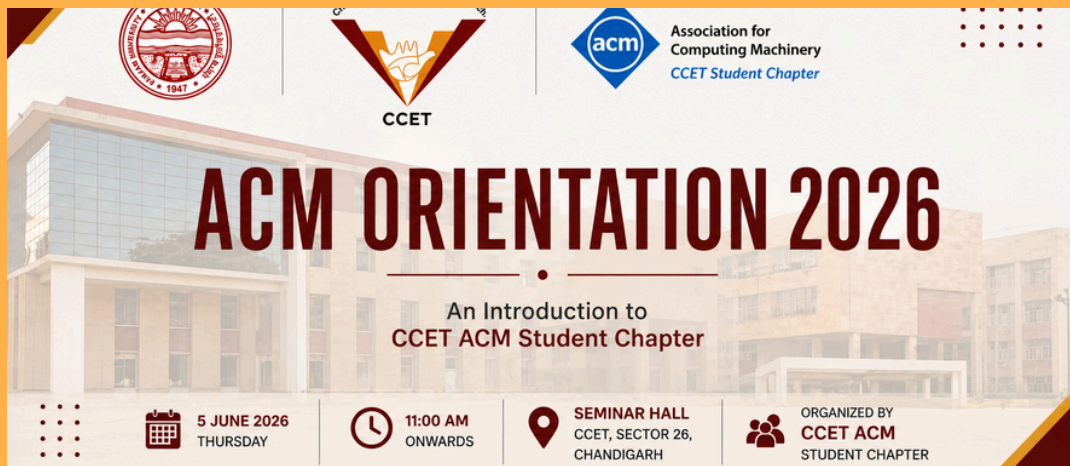
*Authors - Viren Passi, Sudhakar Kumar, Sunil K Singh, Shivam Jindal, Prince Raj, Varsha Arya, Kwok Tai Chui, Brij B Gupta*

## 3. Book Chapters

- **Explainable context-aware IoT framework using human digital twin for healthcare**

*Authors - arun Vats, Sunil K Singh, Sudhakar Kumar, Brij B Gupta, Shabeg Singh Gill, Varsha Arya, Wadee Alhalabi*

# EVENT - 1



1. **CCET ACM Student Chapter held its Orientation Session on 5 June 2026** at Chandigarh College of Engineering and Technology, Chandigarh, for introducing the upcoming junior students to the chapter and its functioning. It provided them with a brief idea about ACM, its functions at CCET, and the various opportunities they can utilize for exploring technology, collaborating with others and developing new skills.
- **Introduction to ACM:** The juniors were introduced to the objectives and functioning of the CCET ACM Student Chapter, along with the areas on which they focus, like technical learning, collaboration, projects, research, workshops, and other student-led activities.
  - **Exploring Opportunities:** The participants learned about the various domains within which they could take part, such as development, cybersecurity, data science, design, research, technical writing, and organizing events.
  - **Life Beyond the Classroom:** The session helped the students understand how ACM could help them get involved in practical project works, attending and organizing various technical events, collaborating with others and building skills over time beyond classroom curriculum.

Thus, the orientation session marked the first step towards introducing the new batch to the CCET ACM community.

# EVENT - 2



2. **The recruitment session for the CCET ACM Student Chapter** took place following the orientation session on **20 June 2026** at Chandigarh College of Engineering and Technology, Chandigarh. The recruitment was done in order to know the personal interests, skills, and potential of the students by the means of an application form along with personal interviews.
- **Exploring Interests:** Students from diverse backgrounds presented a variety of interests, starting from **Cyber Security** and **Machine Learning to Development, Designing, Technical Writing, and Research.**
  - **Interactive Interviews:** The recruitment included interactive interviews, during which the students talked about their interests, skills, projects completed, and achievements, as well as about the areas they wanted to explore.
  - **Building a New Team:** The recruitment was not aimed at identifying the already existing technical skills of the students. Instead, it involved finding out whether they were willing to learn and what ideas they could suggest for working with the chapter.

To conclude, the recruitment drive was the start of building the new team of CCET ACM. With different interests, skills, and ideas, the students laid the foundation for productive cooperation during the coming year.

# Key Insights



# Key Insights



# Brain-Computer Interfaces: Where Cognition meets Code

## ARTICLE

### Abstract

As with so many areas artificial intelligence has broken into in the past few years, none feel more personal than the new frontier AI is reaching today: that of the human skull. ). Since their creation in the 1990s as experimental tools for severely paralyzed patients, BCIs have transformed into what has essentially become an engineering race, with a combination of invasive and non-invasive devices performing the difficult task of accurately translating neural signals into intent with the speed and precision we would only expect out of science fiction a decade ago. [1]

### Introduction

A BCI is fundamentally engineered toward the exact same challenge of transforming a chaotic, multi-dimensional signal into a coherent, precise command. Invasive BCI use devices, called electrode arrays, which are placed directly on or even deep within the brain, to gain highly detailed access to neural cell activity. Companies such as Neuralink, and research projects such as BrainGate, use this sort of approach, where a greater deal of control over neural activity can be acquired. However, invasive BCI can be very expensive and risky due to the surgery required to install electrodes in or on the brain, and also due to concerns of long-term biocompatibility as the devices exist within the body. The most common non-invasive BCI rely on EEG headsets, which record electrical brain activity through the scalp. [1][2][3]

## Ethically Troubling Territory

### 1. Neural Data Has Few Analogues as Far as Sensitivity Data Is Concerned

The unique properties of data generated from an individual's brain are truly second-to-none when one considers personal data, given that these signals have been demonstrated to carry with them information beyond a person's intentional selection. [4]

### 2. Agency And Authenticity

It has been stated by numerous proponents of BCI technology that what they are capable of doing is more often predicting an intent than actually reading a decoded output.

### 3. Dual-Use, coercion risks And "Brainjacking"

The same decoded neural signals that enable the formerly paralyzed to communicate and control complex apparatus could also be used maliciously for surveillance purposes, including coercive interrogation. It has already been empirically proven that 'brainjacking' is a very real concern; hackers can infiltrate a BCI system remotely, without the consent and knowledge of the user, and gain detailed insights into his thoughts,

or even induce malicious behaviors on a controlled device. [6]

## Addressing Concerns

Compared to an incredibly new technology, the field has responded to these issues remarkably quickly, partly because researchers understood from the outset what was ethically at stake.

- **Neurorights Legislation:** Chile has led, amending its constitution to explicitly state protection for "neurorights"- including mental privacy and freedom of thought, and multiple others are following suit. [5]

- **On-Device Processing:** More recent BCIs will decode and interpret neural signals locally to an implanted or head-worn device.

- **Confirmatory Interfaces:** As mentioned previously (see Authorship problem section), many assistive systems rely on user confirmation before any decoded mental task is fully realized. A human must be in the loop throughout the entire predicted action.

- **Institutional Review and Long-Term Consent:** The nature of BCI trials and implants requires long-term implementation- sometimes years of usage with hardware permanently placed in the patient. [4][5]

## Implementation Pathways and Meaningful Work

Brain-Computer Interfaces, are a novel departure from this Replacing/ Tending/ Amplifying framework,

- **Replacing:** For patients who are locked-in, or have severely diminished motor capacity, BCIs can be utilized to restore full communicative capacity and the ability of control lost by physically mediating this functionality with digital ones.

- **Tending the Machine (Hybrid):** Continuous user and scientist interaction, where clinicians are regularly required to update models that decode neural signal input.

- **Amplifying:** Non-clinical, able-bodied volunteers are beginning to be used to test BCIs in the hope of improving task execution- e.g., typing speed, or focus following through enhancement rather than restoration.

## Effect on Various Dimensions

**Task integrity:** For assisted users, successfully inputting their thoughts as output restores task achievement lost in previous iterations of illness- their input into the world is restored.

- **Skill cultivation:** The learning curve of utilizing the interface is itself a skill, and is improving through practice- there is demonstrable improvement in the precision of thought signal delivery with learned practice.

- **Autonomy:** If successful, BCIs can provide independence through direct communication, and not need reliance upon care givers and intermediaries for this basic need.

- **Belongingness:** For non-speaking individuals who may be locked-in, gaining

any method of communication can reestablish crucial bonds with family and their community.

## Risks & Drawbacks

**Signal decay and noise:** Due to the growth of scar tissue around an implanted electrode, [1]

- **Invasive vs. Long-term biocompatibility:**

While not an insurmountable problem, little has yet been written in terms of multi-decade studies on the long term biocompatibility effects of the brain electrode, given the relative newness of the first human prototypes. [6]

- **Security/"Brainjacking":** Any interface or device with digital network access is theoretically accessible by hackers. [6]

- **Access Inequality:** As previously explained.

## Conclusion

Brain-Computer Interfaces represent one of the most direct ways technology has ever attempted to connect with the human body. What once sounded like science fiction is gradually becoming a practical tool, especially for people who have lost the ability to communicate or control physical devices through conventional means. The progress from decoding simple neural signals to controlling cursors, robotic limbs, and communication systems shows just how much potential this field holds.

But the most important part of BCI development is not simply making the technology more powerful. It is making it safe, reliable, accessible, and respectful of the person using it. A system that can decode a neural signal means little if the user cannot

trust where that data goes, how accurately their intentions are being interpreted, or who ultimately controls the technology.

The future of BCI is not necessarily about creating machines that replace human ability. At its best, it is about restoring and extending human agency. If researchers can balance technical ambition with responsible design, BCIs could become one of the most meaningful technologies of the coming decades—not because they make humans more machine-like, but because they can help people regain abilities that they once thought were permanently lost.

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**WRITTEN BY- MANAV JALALI (2024-2028)**



# Explainable AI (X-AI): Opening the Black Box of Machine Learning

## ARTICLE

### Abstract

Artificial intelligence is getting very good at making decisions. But as these systems become more powerful, one question becomes harder to ignore: how do we know why an AI made a particular decision? A model can approve a loan, flag a medical image, or reject an application without giving a reason a person can understand. [3][4]

That is the problem Explainable AI, or XAI, tries to address. It is not about making every model completely transparent. It is about giving people useful ways to understand, question, and evaluate what a model is doing. [3][4]

### Introduction

Modern machine-learning systems can contain millions or even billions of learned parameters. They can recognize patterns that are difficult for a human to describe, but that same complexity makes their decisions difficult to trace. This is the black-box problem. [3][4]

It matters most when AI is used where a wrong decision has real consequences. In healthcare, finance, hiring, and other high-stakes settings, accuracy alone is not always enough. People need to know whether an output makes sense and whether the system is relying on the right information. [3][4]

## So What Is Explainable AI, Really?

Explainable AI is a collection of methods for making machine-learning predictions easier to interpret. Instead of simply accepting an output, XAI asks what factors contributed to it and whether changing important inputs would change the result. [3][4]

Some approaches explain individual predictions, while others show how a model behaves more broadly. The important distinction is that an explanation should be useful, not just visually impressive. A simple explanation that helps someone challenge a wrong prediction can be more valuable than one that only looks convincing. [1][2][3]

## Why This Isn't Optional Anymore

There are three practical reasons explainability has moved from an interesting research topic to something AI systems increasingly need.

- Trust matters. People are more likely to rely on an AI system when they can understand its output and recognize when something looks wrong. [3][4]
- Regulation matters. As governments introduce rules around high-risk AI,

transparency and accountability are becoming part of responsible deployment. [6]

- Debugging matters. When a model gets something wrong, developers need to know why. Explanations can reveal hidden biases, irrelevant features, or patterns the model learned accidentally. [1][3][4]

## The Toolbox: How Engineers Actually Do This

LIME creates a simpler local model around a particular prediction, focusing on what influenced that one decision rather than explaining the entire complex model. [1]

- SHAP assigns importance values to features using ideas from cooperative game theory, providing a consistent way to estimate how much each feature contributed to a prediction. [2]
- Saliency maps and related methods show which parts of an input, such as an image, influenced a model's output. They can help reveal where a computer-vision model was looking. [3][4]
- Attention mechanisms can provide clues about which inputs a model considered important, although attention should not automatically be treated as a complete explanation of reasoning. [3][4]
- Counterfactual explanations ask what would need to change for the model to produce a different outcome. This can make explanations more intuitive because they connect directly to possible alternatives. [5]

## Where This Actually Shows Up in the Real World

In healthcare, explainability can help clinicians examine why an AI system flagged a scan or predicted a particular risk. In finance, it can help explain credit decisions. In hiring, it can expose whether a system is relying on inappropriate patterns. The goal is not to replace human judgment, but to help people evaluate the machine's output. [3][4][6]

Explainability is also useful during development. When engineers can see what a model is responding to, they can identify unexpected correlations and investigate errors before deployment. XAI can therefore serve both users and developers. [1][2][3]

The biggest trade-off is accuracy versus interpretability. Simpler models are often easier to understand, while highly complex models can capture patterns that simpler systems miss. [3][4]

- Post-hoc explanations can also be misleading. An explanation generated after a decision may look reasonable without necessarily describing the model's actual internal reasoning. [3][4]
- There is also no single definition of a 'good' explanation. A developer, regulator, doctor, and ordinary user may need very different kinds of explanations. [3][4]

## Where This Is All Heading

The next stage of XAI is moving beyond simply showing which features mattered toward explanations that are more useful

for reasoning about cause and effect. Researchers are also exploring counterfactual and interactive explanations that let users ask what would happen if a factor changed. [3][5]

## Conclusion

The real value of XAI is not that it makes AI infallible. It gives people a way to interact with uncertainty. A model can still be wrong, but an understandable system gives users a better chance of spotting that mistake and deciding what to do next. [3][4]

- A useful explanation should help a person understand the decision rather than simply persuade them to accept it. [3][4]
- Different users need different levels of detail. A technical explanation for an engineer will not necessarily be useful to a doctor, regulator, or everyday user. [3][4]
- Explainability works best alongside testing, documentation, human oversight, and responsible deployment rather than as a substitute for them. [3][6]

As AI takes on more responsibility for decisions that genuinely affect people's lives, explaining those decisions is becoming almost as important as getting them right. XAI cannot make every model transparent, but it can make complex systems easier to question, audit, debug, and use responsibly. [3][4][6]

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**WRITTEN BY- SNEHA THAKUR (2024-2028)**



# Affective Computing: Multimodal Emotion Inference in Human-Centered AI Systems

## ABSTRACT

Affective Computing is a branch of science that aims to enable computers to recognise, understand and react to emotions of people. Combining various signals such as facial expressions, voice, language and physiological information Affective Computing attempts to improve interaction between humans and computers. However emotions are complex entities. Influenced by many factors including culture and context making inference about emotions is not an easy task. The article examines mechanisms of emotion recognition and discusses possible issues regarding accuracy, consent, privacy and behavioural targeting. Also some potential uses of Affective Computing are examined along with possible ethical guidelines. It is important to create emotion-aware technology respecting human autonomy and privacy.

## Introduction

A phone that senses stress before a word is typed, a car that flags driver drowsiness from blink rate, [1][5]

The field traces back to Rosalind Picard's 1997 formulation at MIT, but its practical maturity is recent. Early systems were bottlenecked by weak computer vision and coarse sensors; the gap between "frowning because annoyed" and "frowning because concentrating" was simply too fine-grained for the models of the time to resolve reliably. [1]

## ARTICLE

### Modalities of Emotion Inference

Emotion recognition covers several distinct signal-processing pipelines, often combined for robustness. [5][6]

- Facial expression analysis uses convolutional networks trained to map facial action units (muscle movement patterns catalogued by the Facial Action Coding System, [6])
- Vocal prosody analysis extracts paralinguistic features independent of lexical content: pitch contour, speech rate, energy, pause duration. [6]
- Physiological measurement captures autonomic nervous system activity: heart rate variability (HRV), galvanic skin response (GSR), and pupillometry. [5]
- Text-based sentiment and emotion classification infers affective state from written language using models ranging from lexicon-based scoring to fine-tuned transformer classifiers. It is the cheapest modality to collect at scale and generally the least reliable in isolation. [6]
- Multimodal fusion architectures combine two or more of these channels, typically through early fusion (concatenating features before classification) or late fusion (combining independent per-modality predictions), [5][6]

# Conceptual and Ethical Challenges

## The Universality Assumption

Most commercial systems are still built on Paul Ekman's model of six to seven "basic emotions" expressed consistently across cultures. [2][3]

- This assumption has been substantially challenged. [3]
- Systems trained predominantly on Western facial datasets show measurable accuracy degradation on populations with different expressive norms, a gap documented in cross-cultural benchmarking studies rather than a purely theoretical concern. [3]

## Consent and Data Sensitivity

- Emotion inference in classrooms, workplaces, and public spaces frequently occurs without explicit, informed opt-in: an "engagement-tracking" camera in a lecture hall is a common example. [4]
- Even where consent is nominally obtained, affective data is difficult to meaningfully anonymize, since the underlying signal (a face, a voice) is itself identifying.

## Inferred State and Behavioral Targeting

- Advertising and recommendation systems have a direct incentive to detect emotionally salient moments, such as boredom, loneliness, or anxiety, and time content accordingly.
- The distinction between adaptive, user-beneficial personalization and exploitation of a detected emotional vulnerability is a matter of intent and disclosure, not of the underlying technique, which makes it difficult to regulate at the model level.

# Mitigation and Governance Approaches

- Regulatory constraint: The EU AI Act designates emotion-recognition systems as high-risk (in most cases prohibited outright) when deployed in workplace and educational settings. It is the most concrete legal response to affective computing to date.
- Uncertainty-calibrated outputs: Rather than emitting a single confident label, newer systems increasingly output confidence intervals or probability distributions over candidate emotional states, reducing the risk of overconfident misclassification driving downstream decisions. [5]
- Cross-cultural dataset diversification: Training corpora are gradually incorporating broader demographic and cultural representation, though benchmark datasets remain disproportionately Western-sourced. [5]
- Purpose-limited, opt-in deployment: Restricting emotion inference to contexts where users explicitly consent to the specific use case (a therapeutic app, for instance) is increasingly treated as the more defensible design pattern relative to ambient, always-on monitoring.

# Impact on Dimensions of Meaningful Work

- Task Integrity: In therapeutic contexts, continuous affect monitoring can help ensure sessions address a client's actual state rather than only their self-report.

- **Skill Cultivation:** Sustained reliance on automated emotion detection carries a documented risk of atrophying human interpretive skill, a concern raised specifically around classroom deployment.
- **Autonomy:** Retained control over who accesses one's affective data, and the ability to opt out without penalty, is the central autonomy question in this domain.
- **Belongingness:** Appropriately deployed, affect-aware assistive technology can improve perceived understanding for users who struggle with self-expression; inappropriately deployed, it produces a surveillance effect rather than a supportive one.

## Limitations and Risks

- **Bounded Accuracy.** Even state-of-the-art systems struggle with mixed, subtle, or culturally atypical emotional expression, and performance drops measurably outside training-distribution demographics. [3][5]
- **Correlation Without Comprehension.** Detection systems learn statistical association between observable signal and label, which is distinct from any model of the underlying affective or cognitive process, a distinction frequently elided in commercial framing. [3]

## Conclusion

Affective computing occupies a genuinely useful niche in narrow, consensual applications such as accessibility tools, therapeutic support, and safety-critical drowsiness detection, while raising substantive concern the moment inference occurs without disclosure.

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**WRITTEN BY- SUNDRAM TIWARI (2024-2028)**



# Data-Driven Cybersecurity Framework for Smart Enterprise Systems

## Abstract

Modern enterprises are becoming increasingly connected through cloud platforms, Internet of Things (IoT) devices, and industrial automation. That connectivity brings efficiency, but it also creates more places for attackers to enter. Data-driven cybersecurity responds to this challenge by combining large-scale data analysis, artificial intelligence, machine learning, and real-time monitoring to detect threats earlier and respond faster. This article looks at how these technologies work together, where they are used, and the practical challenges involved in building security systems that can learn and adapt as threats change. [1]

## Keywords

Big Data Analytics, Machine Learning, Intrusion Detection Systems (IDS), Security Information and Event Management (SIEM), Cyber-Physical Systems Security, Predictive Threat Intelligence

## Introduction

Enterprise networks no longer have a clear boundary. A single organization may depend on cloud services, connected devices, remote endpoints, and industrial systems at the same time. While this makes operations faster and more scalable, it also increases the number of paths through which an attacker can reach the system. [1]

Traditional security tools often rely on predefined rules and known signatures. They remain useful, but they can struggle when an attack looks different from anything seen before. Data-driven cybersecurity takes a different approach: it continuously collects security data, analyzes patterns, and uses machine-learning models to identify unusual behavior. Instead of only reacting after an incident, the aim is to recognize warning signs early enough to prevent or limit the damage. [5]

## Data-Driven Cybersecurity Technologies

At the center of a data-driven security framework is a large stream of information coming from many parts of an enterprise. Network logs, endpoints, IoT devices, cloud services, and other systems all contribute pieces of the security picture. Before this information can be analyzed, it needs to be collected, cleaned, normalized, and stored in a form that security tools can work with. [1][6]

Machine learning and artificial intelligence then help identify patterns that may not be obvious to a human analyst.

A model can learn what normal activity looks like and flag deviations that deserve attention. Security Information and Event Management (SIEM) systems add another layer by bringing security events together in one place, while Security Orchestration, Automation and Response (SOAR) platforms can automate parts of the response. [2]

Cloud and edge computing also matter because security data increasingly comes from distributed environments. Together, these technologies move cybersecurity toward a continuous cycle of monitoring, analysis, decision-making, and response rather than a purely reactive model. [1]

## The Role of Data-Driven Cybersecurity in Smart Enterprises

The more connected an enterprise becomes, the harder it is to understand what is happening across the entire system. Data-driven security helps by bringing information from different sources together and looking for relationships between events. An unusual login, unexpected network traffic, and abnormal activity on an IoT device might look harmless individually, but together they can form a much stronger warning signal. [5]

Another advantage is the ability to look forward rather than only backward. Machine-learning systems can identify anomalies and use patterns from previous activity to estimate where vulnerabilities or attacks may develop.

This is particularly valuable in environments such as healthcare, finance, and smart manufacturing, where a security incident can affect both sensitive information and physical operations. [4]

## Cyber Threat Detection and Response Lifecycle

Data-driven cybersecurity works best as a continuous loop rather than a one-time process. It begins with monitoring, where information is collected from networks, devices, and applications. The collected data is then analyzed to identify anomalies and determine whether they represent a genuine threat. [2]

- Monitor: continuously collect security information from networks, endpoints, cloud systems, and connected devices.
- Detect and analyze: identify unusual behavior and examine its possible cause, severity, and impact.
- Respond: contain the threat through actions such as blocking malicious activity or isolating a compromised system.
- Recover and learn: restore normal operation and feed lessons from the incident back into the security process.

The feedback stage is especially important. Every incident can provide new information that improves future detection, allowing the system to gradually adapt instead of repeatedly facing the same type of threat as if it were seeing it for the first time. [2][5]

# Machine Learning-Based Intrusion Detection System

Machine-learning-based Intrusion Detection Systems (IDS) are an important part of this approach. They examine network traffic and system logs for patterns that may indicate suspicious behavior. The process generally begins with data collection, followed by feature extraction, where useful characteristics such as traffic flow, protocols, or other system attributes are selected. [5]

These features are used to train a model to distinguish normal activity from abnormal behavior. Once trained, the model can examine new information in real time and classify it accordingly. When the system detects something suspicious, it can generate an alert or trigger a corrective action.

The major advantage is adaptability. Instead of depending entirely on a fixed list of known attack signatures, machine-learning systems can learn patterns from data and potentially identify unusual activity that does not exactly match previously defined rules. That makes them particularly useful in environments where the threat landscape changes quickly. [5]

## Challenges and Limitations

The idea sounds straightforward, but building a reliable data-driven security system is not. The first challenge is scale. Enterprises can generate enormous amounts of security data from many different sources, and processing all of it in real time requires substantial

computing and storage resources. [6]

Machine learning also introduces its own uncertainty. A model can produce false positives, miss an attack, or behave differently when it encounters data that is unlike its training data. Complex models can also be difficult to explain, which becomes a problem when security teams need to understand why a particular alert was generated. [3]

Cost is another practical barrier. Effective data-driven cybersecurity requires infrastructure, specialized expertise, continuous maintenance, and regular model updates. Finally, collecting and analyzing large amounts of user and system data creates privacy concerns. Strong security therefore has to be balanced with responsible data handling and appropriate access controls. [4]

## Conclusion

Data-driven cybersecurity represents a shift in how modern enterprises approach security. Instead of waiting for a known attack and then reacting to it, organizations can continuously collect information, identify unusual behavior, assess risk, and automate parts of the response. Big data analytics, machine learning, SIEM, and automated security platforms each contribute a different piece to this process.

At the same time, the technology is not a magic shield. Data quality, computing requirements, false alarms, explainability, cost, and privacy can all limit its effectiveness. The real value comes from building a system that combines intelligent automation with human oversight and continuous learning.

As enterprise environments become more connected, this ability to adapt may become one of the most important parts of keeping them secure. [3][4]

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**WRITTEN BY- DEVESH VAIDYA (2024-2028)**



# The Rise of TinyML: AI on Edge Devices

## ARTICLE

### Abstract

TinyML brings machine learning directly to small, low-power devices such as microcontrollers and sensors. Instead of sending every piece of data to the cloud, these systems can process information locally, enabling faster decisions, lower bandwidth use, and better privacy. This article explores how TinyML works, where it is being used, the tools that support it, and the challenges involved in bringing intelligent models onto hardware with very limited resources. [1][3]

### Introduction

AI and machine learning have transformed areas ranging from healthcare and agriculture to manufacturing and transportation. Much of this intelligence, however, still depends on cloud servers and a reliable internet connection. That can introduce latency, consume bandwidth, increase costs, and raise privacy concerns when sensitive data has to leave the device. [1][3]

TinyML approaches the problem from the other direction: instead of moving data to powerful machines, it brings a small amount of intelligence to the device itself. A smartwatch, environmental sensor, or industrial controller can run an ML model locally, making TinyML useful where power, connectivity, and physical space are limited. [1][3]

### Understanding TinyML

TinyML is more than simply shrinking a machine-learning model. It is an entire workflow built around extremely limited hardware. Models are normally trained on a PC or cloud environment and then optimized before being deployed on microcontrollers that may have only kilobytes of RAM and modest processing power. [1]

The field brings together embedded systems, machine learning, model compression, and hardware-aware optimization. The result is AI designed to work within the constraints of the device rather than expecting the device to provide cloud-level resources. [1][2]

The process begins much like conventional machine learning: a model is trained on a large dataset using powerful hardware. Once it performs well enough, it has to be made small and efficient enough to run on the target device. [1][2]

- Quantization reduces the precision of model parameters, lowering memory and computation requirements. [2]
- Pruning removes less important connections, reducing model size while aiming to preserve useful accuracy. [2]
- Knowledge distillation can transfer useful behavior from a larger model into a smaller one. [2]

The optimized model is then deployed using frameworks such as TensorFlow Lite for Microcontrollers. Inference takes place directly on sensor data, allowing the device to make decisions without constantly communicating with a cloud server. [3][4]

## Applications of TinyML

**Healthcare:** Wearable devices can analyze signals such as heart activity or movement locally and provide rapid alerts without continuously sending personal data to the cloud. [1][3]

• **Agriculture:** Embedded models can process soil and crop sensor data, helping detect conditions that require attention while reducing dependence on connectivity. [3]

• **Industrial IoT:** Vibration and temperature data can be analyzed locally to identify signs of equipment problems and support predictive maintenance. [1][3]

• **Smart Homes:** Motion detection and voice-triggered functions can respond locally, improving responsiveness while reducing unnecessary data sharing. [3][4]

• **Environmental Monitoring:** Low-power sensor nodes can operate in remote areas to monitor air quality, fires, or other conditions with minimal energy use. [3]

## Tools and Frameworks

TensorFlow Lite for Microcontrollers provides tools for running ML inference on microcontrollers without requiring a full operating system. [3][4]

• Edge Impulse supports data collection, model development, optimization, and deployment through an integrated workflow.

• CMSIS-NN provides optimized neural-network kernels for ARM Cortex-M processors.

• Arduino boards such as the Nano 33 BLE Sense make TinyML experimentation accessible for prototyping and education.

• microTVM and similar tools help compile and execute models with a small runtime footprint.

## Challenges

• **Accuracy vs. Size:** Making a model smaller can affect accuracy, so developers have to balance performance against resource use. [1][2]

• **Hardware Heterogeneity:** Different microcontroller architectures require different optimization strategies, making deployment less straightforward.

• **Security:** Edge devices can be physically accessible, creating risks such as tampering and unauthorized access.

• **Model Updates:** Devices deployed in the field may not have reliable mechanisms for receiving updated models.

• **Debugging:** Limited memory, interfaces, and processing resources can make diagnosing problems more difficult.

## Key Features of TinyML

• **Low Latency:** Local inference removes the communication delay involved in sending data to a remote server.

• **Privacy Preservation:** Sensitive information can remain on the device instead of being transmitted for processing.

• **Energy Efficiency:** Optimized models can operate within the power budgets of battery-powered devices.

## Future Directions

The next stage of TinyML is likely to move beyond simply running a trained model on a small device. Researchers are exploring ways for devices to adapt locally, while federated learning could allow multiple devices to improve models collectively without directly sharing raw data. [1][3]

Energy harvesting is another promising direction. If devices can draw power from light, heat, or movement, TinyML systems could remain active for long periods in remote locations. Specialized low-power hardware and more standardized frameworks could also make deployment easier across different platforms. [1][3][4]

## Conclusion

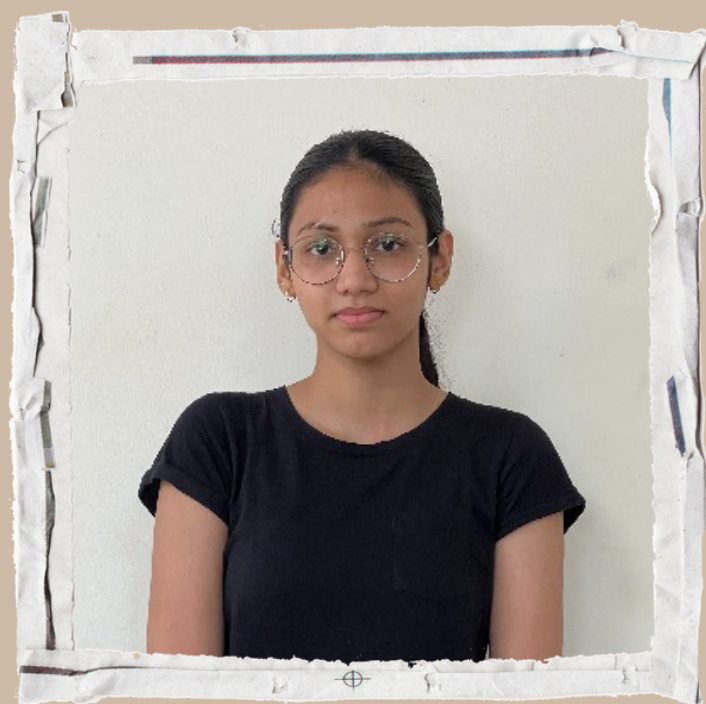
TinyML changes the way we think about where artificial intelligence has to live. Instead of depending entirely on large servers, useful intelligence can be placed directly inside the small devices already surrounding us. This makes real-time, private, and offline decision-making possible where sending data to the cloud is slow, expensive, or impractical. [1][3]

There are still real limitations. Developers have to work within tight memory and power budgets, balance model accuracy against size, and deal with hardware, security, and update challenges. Even so, the technology is moving quickly. As edge hardware and optimization techniques improve, TinyML could make intelligent computing increasingly invisible—embedded directly into the devices that quietly collect and respond to information around us. [1][3][4]

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WRITTEN BY- BHUMIKA BIJWAN (2023-2027)



# Climate-Conscious Computing: Creating Algorithms with the Environment in Mind

## ARTICLE

### Abstract

Computing is getting more powerful, but it is also using more energy. As data centers, AI systems, and digital services continue to grow, their environmental impact is becoming harder to ignore [1][2]. Climate-conscious computing asks a simple question: can we design algorithms that are not only fast and accurate, but also more responsible with energy and resources? This article looks at practical ways to make computing more sustainable and why that matters.

### Introduction

Computers are now part of almost everything we do, from communication and banking to healthcare and scientific research. But behind these services are data centers, networks, and increasingly large AI models that require significant amounts of energy. The digital revolution has brought enormous benefits, but its environmental cost is becoming part of the conversation [1][2][3].

This is where climate-conscious computing comes in. Instead of treating sustainability as something to think about after a system is built, it makes environmental impact part of the design itself. An algorithm should ideally be judged not only by how quickly or accurately it works, but also by how much unnecessary computation it requires [1][3].

### The Computing Energy Cost

Modern computing infrastructure operates at a huge scale. Data centers already consume a significant share of global electricity, while demand from AI and other computation-heavy applications continues to grow [2]. Even though processors and hardware are becoming more efficient, the amount of computation we ask them to perform is growing just as quickly, or faster.

That means improving individual hardware components is not enough. We also need to ask whether the algorithms themselves are doing more work than necessary. Energy and carbon cost can become another design constraint, alongside time, memory, and accuracy [1][2].

Climate-aware computing means including environmental impact when making technical decisions. A developer might choose a smaller model, reduce unnecessary data movement, use fewer computation cycles, or schedule workloads when cleaner energy is available [1][2].

The idea is not to make every system less capable. It is to find cases where a little less computation can provide almost the same result. If two approaches solve the same problem, but one uses considerably fewer resources, there is a good reason to consider the more efficient option [3][4].

## Climate-Aware Computing: Design & Applications

- **Carbon Cost as a Metric:** Energy use and carbon emissions can be considered alongside traditional measures such as time and memory [1][3].
- **Adaptive Accuracy:** Some applications can tolerate small reductions in precision. Early stopping and reduced-precision computation can lower unnecessary work.
- **Reduce Data Movement:** Moving information between memory, processors, and networks also consumes energy. Better data locality and compression can help [2].
- **Green Scheduling:** Workloads can be scheduled according to available infrastructure and periods when renewable energy is more accessible [2][4].
- **Think About the Full Lifecycle:** Sustainability should be considered during training, deployment, maintenance, updates, and eventual retirement of a system [1][3].
- **Machine Learning:** Techniques such as pruning, quantization, and knowledge distillation can make models smaller and reduce the computation needed during inference [3].
- **Edge Computing:** Processing data closer to where it is generated can reduce repeated communication with cloud servers and lower transmission overhead [2][4].
- **Environmental Modelling:** Weather, climate, and ecosystem simulations can benefit from balancing useful accuracy with the amount of computation required [4].

## Case Study: Google Data Centers

One well-known example comes from DeepMind's work with Google on data-center cooling. An AI-based system used information such as sensor readings, weather conditions, and expected workloads to recommend changes to cooling operations [3][4].

The example shows that algorithms designed around changing conditions can reduce resource use without simply reducing performance. Instead of treating energy consumption as fixed, the system used prediction and adaptive control to make the infrastructure itself more efficient [3][4].

## Challenges, Responsibility & Future Directions

Sustainability is not only an engineering problem. Decisions about model size, training methods, hardware, and deployment all have consequences beyond the computer itself. Environmental impact therefore belongs alongside other responsible-AI concerns such as fairness, safety, and transparency [1][3].

This does not mean choosing the smallest or least powerful solution every time. The better question is whether the extra computation is actually providing enough value to justify its cost. Making that trade-off visible is an important part of responsible design [1][4].

## Conclusion

Climate-conscious computing asks us to expand the way we think about efficiency. Speed, accuracy, and scalability still matter, but so do energy use and environmental impact. By reducing unnecessary computation and data movement, choosing appropriate models, and considering sustainability throughout a system's lifecycle, developers can make computing more responsible without giving up its benefits [1][2][3].

The goal is not to slow technological progress. It is to make that progress more thoughtful. As digital systems become a larger part of everyday life, designing them with the planet in mind may become just as important as designing them for the people who use them [3][4].

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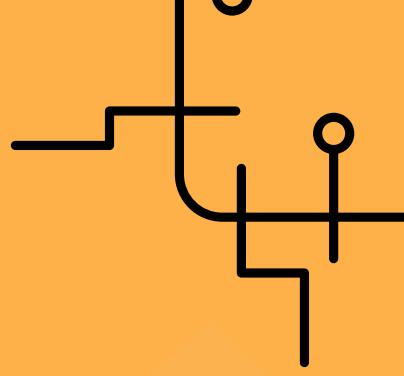
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# FUN TECH FACTS

- **The First Computer Bug Was Actually a Bug:** In 1947, engineers working on the Harvard Mark II discovered a moth trapped inside one of its relays, causing the machine to malfunction. They taped the moth into their logbook and called it a “bug,” helping popularize the term “debugging” in computing.
- **The First Computer Mouse Was Made of Wood:** In 1964, Douglas Engelbart’s team created the first computer mouse prototype as a small wooden box with two wheels underneath. The wheels tracked movement along the X and Y axes, forming the basic idea behind the computer mice we use today.
- **The First Webcam Watched a Coffee Pot:** In 1991, researchers at the University of Cambridge connected a camera to their network and pointed it at a coffee pot. The live feed allowed people to check if coffee was available without leaving their desks, making it one of the earliest uses of a webcam.
- **The First Website Is Still Online:** Created at CERN in 1991, the world’s first website explained what the World Wide Web was and how people could use it. Unlike most early websites that disappeared, the original page has been restored and remains accessible today.

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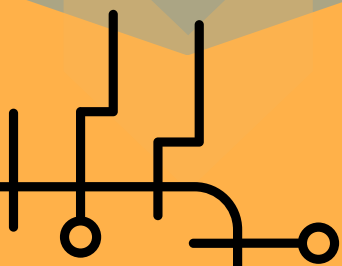
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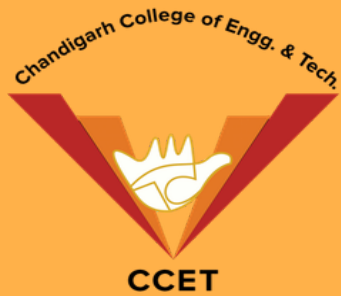
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“Technology is best when  
it brings people together.

—Matt Mullenweg

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